

Q1.

```
class QueueNode {
    QueueNode next;
    int data;
    //constructor

    QueueNode(int d){
        this.data = d;
    }
}
class Queue{

    QueueNode front;
    QueueNode rear;
    // Enqueue Function

    void enqueue(int data){
        QueueNode newNode = new QueueNode(data);

        if(this.front == null){
            this.front = this.rear = newNode;
            return;
        }else{
            rear.next = newNode;
            rear = newNode;
        }
    }

    //Dequeue Function

    void dequeue(){
        if(this.front == null){
            return;
        }
        QueueNode temp = this.front;
        System.out.println(this.front.data);
        this.front = this.front.next;

        if(this.front== null){
            this.rear=null;
        }
    }
}
```

```

// peek

void peek(){

    if(this.front == null){
        System.out.println("the queue is empty");
    }
    else{
        System.out.print(this.front.data);
    }
}

// Display

void display(){
    QueueNode temp = this.front;
    while(temp != null){
        System.out.println(temp.data);
        temp = temp.next;
    }

}

}

```

```

class Driver{
    public static void main( String []args){

        Queue q = new Queue();
        //1
        q.enqueue(2);
        q.enqueue(5);
        q.enqueue(38);
        q.enqueue(44);
        q.enqueue(73);
        q.enqueue(105);
        q.enqueue(225);
        q.enqueue(515);
    }
}

```

```
//2
```

```
q.dequeue();
```

```
//3
```

```
q.enqueue(30);
```

```
q.enqueue(10);
```

```
//4
```

```
q.dequeue();
```

```
q.enqueue(838);
```

```
q.enqueue(586);
```

```
//5
```

```
q.display();
```

```
}
```

```
}
```

Q2.

```
//VEHICLE
```

```
class Vehicle{
```

```

        public String VIN;
        public String vType;
        public String serviceType;

        public Vehicle(String vin, String vt, String st){

            this.VIN= vin;
            this.vType= vt;
            this.serviceType= st;

        }
    }
}

```

```

class QueueNode {
    QueueNode next;
    Vehicle data;
    //constructor

    QueueNode(Vehicle d){
        this.data = d;
    }
}

```

```

class Queue{

    QueueNode front;
    QueueNode rear;

    //inLine

    void inLine(String VIN){
        QueueNode temp = this.front;
        int i=0;

        String vin = temp.data.VIN;

        while((vin.equals(VIN)!=true)){

```

```

        i++;
        temp = temp.next;
        vin =temp.data.VIN;

    }
    System.out.println("Number of vehicles to be serviced : " + i);
}

```

//enterForService

```

void enterForService(String VIN, String Vt,String St){

    Vehicle vehicle = new Vehicle(VIN,Vt,St);

    enqueue(vehicle);

}

```

//exit

```

void exitService(){
    if(this.front == null){
        return;
    }
    QueueNode temp = this.front;

    this.front = this.front.next;

    if(this.front== null){
        this.rear=null;
    }
}

```

// Enqueue Function

```

void enqueue(Vehicle data){
    QueueNode newNode = new QueueNode(data);

    if(this.front == null){
        this.front = this.rear = newNode;
        return;
    }else{
        rear.next = newNode;
        rear = newNode;
    }
}

```

// Display

```

void showQueue(){
    QueueNode temp = this.front;
    while(temp != null){
        Vehicle v = temp.data;

        System.out.println(" Vehicle Identification No: " + v.VIN);
        System.out.println(" Vehicle type : " + v.vType);
        System.out.println(" Service type : " + v.serviceType);

        temp = temp.next;
    }
}

```

```

class Driver{
    public static void main( String []args){

```

```
Queue q = new Queue();
```

```
q.enterForService("AB200","SUV","oil change");  
q.enterForService("CS250","Van","Body wash");  
q.enterForService("LK560","Jeep","full service");  
q.enterForService("NM301","Car","oil change");  
q.enterForService("LK900","SUV","Interior cleaning");
```

```
//q.showQueue();
```

```
//q.showQueue();
```

```
// q.exitService();
```

```
// q.showQueue();
```

```
q.inLine("LK560");
```

```
q.showQueue();  
q.exitService();
```

```
System.out.println("++++");
```

```
q.showQueue();
```

```
}
```

```
}
```